

## SolarLink 7 — Team Competence Analysis

Competence areas mapped against assigned project workstreams. X = lead competence; \* = supporting competence.

| Member               | Site | Role         | Project Management | CAD / 3D Design | 3D Printing | Hardware Assembly | Firmware (Arduino) | MQTT / IoT | Node-RED / Dashboard | Documentaion |
|----------------------|------|--------------|--------------------|-----------------|-------------|-------------------|--------------------|------------|----------------------|--------------|
| Swayam Jakhalekar    | THM  | Communicator | X                  |                 |             |                   | X                  | X          | X                    | X            |
| Shanmugaraja Senthil | THM  | Organiser    | •                  | X               |             | X                 |                    |            |                      | X            |
| Shantanu Shende      | THM  | Member       |                    |                 | X           | •                 | X                  | X          | X                    | •            |
| Gabriel Lenoir       | CESI | Communicator | X                  |                 |             |                   | •                  | X          | X                    | X            |
| Willyam Okondza      | CESI | Organiser    | •                  | X               | •           | X                 | •                  |            |                      | •            |
| Edouard Cossin       | CESI | Member       |                    |                 | X           | X                 |                    |            |                      | X            |

|                |
|----------------|
| <b>Legend:</b> |
|----------------|

*X = lead competence • • = supporting competence • blank = not engaged on this workstream*

## SolarLink 7 — Distribution of Tasks Between Locations

Mapping of project tasks to ownership type and location. Site-local = single-site task; Joint = co-owned cross-site; Cross-site coordination = communicator-led bridging task.

| WBS ID | Task                                   | Type                    | THM Owner    | CESI Owner | Notes                                              |
|--------|----------------------------------------|-------------------------|--------------|------------|----------------------------------------------------|
| 1.1    | Planning document & baseline           | Cross-site coordination | Swayam       | Gabriel    | Communicators jointly own document                 |
| 1.2    | Weekly steering & minutes              | Cross-site coordination | Swayam       | Gabriel    | Thursday 12:00 CET, Google Meet                    |
| 1.3    | Risk & issue log                       | Cross-site coordination | Swayam       | Gabriel    | Living document, GitHub-tracked                    |
| 1.4    | Report — technical sections            | Joint                   | Shanmugaraja | Edouard    | Firmware, electronics, mechanical                  |
| 1.5    | Report — PM sections                   | Joint                   | Swayam       | Gabriel    | WBS/Gantt retrospective, lessons learned           |
| 1.6    | Report — analysis & conclusions        | Joint                   | All THM      | All CESI   | Whole-team contribution                            |
| 1.7    | Final report assembly & review         | Cross-site coordination | Swayam       | Gabriel    | Communicators integrate sections                   |
| 2.1    | Static panel holder — TinkerCAD design | Joint                   | Shanmugaraja | Willyam    | Lead designer: Willyam; handover to THM via GitHub |
| 2.2    | Print static holder — THM              | Site-local              | Shantanu     | —          | Local print at THM open.A5 Makerspace              |
| 2.3    | Print static holder — CESI             | Site-local              | —            | Edouard    | Local print at CESI facility                       |
| 2.4    | SPP CAD design                         | Joint                   | Shanmugaraja | Willyam    | Lead designer: Willyam                             |
| 2.5    | Print SPP brackets — THM               | Site-local              | Shantanu     | —          | Local print at THM                                 |
| 2.6    | Print SPP brackets — CESI              | Site-local              | —            | Edouard    | Local print at CESI                                |
| 2.7    | Servo mounting & calibration           | Joint                   | Shanmugaraja | Willyam    | Project week                                       |
| 3.1    | BOM & wiring schematic                 | Joint                   | Shanmugaraja | Willyam    | Single shared schematic for both sites             |
| 3.2    | Build & bring-up — THM                 | Site-local              | Shanmugaraja | —          | Identical kit, parallel build                      |
| 3.3    | Build & bring-up — CESI                | Site-local              | —            | Willyam    | Identical kit, parallel build                      |
| 3.4    | INA219 calibration                     | Joint                   | Swayam       | Gabriel    | Cross-validate with multimeter                     |
| 3.5    | Sensor + servo firmware                | Joint                   | Swayam       | Gabriel    | Lead: Swayam; handover to Gabriel                  |
| 3.6    | Tracking algorithm                     | Joint                   | Swayam       | Gabriel    | Lead: Swayam; time-based, NTP + sun position       |
| 3.7    | Local LCD display                      | Joint                   | Shanmugaraja | Willyam    | Optional; voltage/current/power output             |
| 4.1    | MQTT topic schema (cesithmcoil2025/)   | Joint                   | Swayam       | Gabriel    | Co-designed by communicators                       |
| 4.2    | MQTT client on Arduino                 | Joint                   | Shantanu     | Willyam    | Setup lead: Shantanu; CESI support                 |
| 4.3    | Node-RED ingest flow                   | Joint                   | Shantanu     | Edouard    | Setup lead: Shantanu                               |
| 4.4    | Node-RED VPP dashboard                 | Joint                   | Shantanu     | Edouard    | Setup lead: Shantanu                               |
| 4.5    | Remote-command handler                 | Joint                   | Shantanu     | Edouard    | End-to-end bidirectional control                   |
| 5.1    | Site-local end-to-end test             | Site-local              | All THM      | All CESI   | Each site validates own prototype                  |
| 5.2    | Cross-site MQTT test                   | Cross-site coordination | All THM      | All CESI   | Both sites publish simultaneously                  |
| 5.3    | IV-curve measurements                  | Site-local              | All THM      | All CESI   | Multimeter-based; 27 May project week              |
| 5.4    | MPP + optimal load resistor            | Site-local              | All THM      | All CESI   | Derived from IV-curve data                         |
| 5.5    | 30-min energy measurement              | Site-local              | All THM      | All CESI   | Fixed tilt, optimal load                           |
| 5.6    | Project week integration               | Cross-site coordination | All THM      | All CESI   | 27–29 May, joint working window                    |

## SUMMARY

|                               |    |
|-------------------------------|----|
| Site-local tasks              | 10 |
| Joint tasks                   | 16 |
| Cross-site coordination tasks | 6  |
| Total tasks                   | 32 |